Sharma_DSC520Week10.3FinalStep2

2025-02-10

```
churn <- read.csv('Telco-Customer-Churn.csv')
str(churn)
```

```
## 'data.frame':    7043 obs. of  21 variables:
##  $ customerID      : chr  "7590-VHVEG" "5575-GNVDE" "3668-QPYBK" "7795-CFOCW" ...
##  $ gender          : chr  "Female" "Male" "Male" "Male" ...
##  $ SeniorCitizen   : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ Partner         : chr  "Yes" "No" "No" "No" ...
##  $ Dependents      : chr  "No" "No" "No" "No" ...
##  $ tenure          : int  1 34 2 45 2 8 22 10 28 62 ...
##  $ PhoneService    : chr  "No" "Yes" "Yes" "No" ...
##  $ MultipleLines   : chr  "No phone service" "No" "No" "No phone service" ...
##  $ InternetService : chr  "DSL" "DSL" "DSL" "DSL" ...
##  $ OnlineSecurity  : chr  "No" "Yes" "Yes" "Yes" ...
##  $ OnlineBackup    : chr  "Yes" "No" "Yes" "No" ...
##  $ DeviceProtection: chr  "No" "Yes" "No" "Yes" ...
##  $ TechSupport     : chr  "No" "No" "No" "Yes" ...
##  $ StreamingTV     : chr  "No" "No" "No" "No" ...
##  $ StreamingMovies : chr  "No" "No" "No" "No" ...
##  $ Contract        : chr  "Month-to-month" "One year" "Month-to-month" "One year" ...
##  $ PaperlessBilling: chr  "Yes" "No" "Yes" "No" ...
##  $ PaymentMethod   : chr  "Electronic check" "Mailed check" "Mailed check" "Bank transfer (automatic)
##  $ MonthlyCharges  : num  29.9 57 53.9 42.3 70.7 ...
##  $ TotalCharges    : num  29.9 1889.5 108.2 1840.8 151.7 ...
##  $ Churn           : chr  "No" "No" "Yes" "No" ...
```

```
sapply(churn, function(x) sum(is.na(x)))
```

```
##       customerID           gender    SeniorCitizen          Partner
##                0                0                0                0
##       Dependents           tenure     PhoneService    MultipleLines
##                0                0                0                0
##  InternetService   OnlineSecurity     OnlineBackup DeviceProtection
##                0                0                0                0
##      TechSupport      StreamingTV  StreamingMovies         Contract
##                0                0                0                0
## PaperlessBilling    PaymentMethod   MonthlyCharges     TotalCharges
##                0                0                0               11
##            Churn
##                0
```

```
churn <- churn[complete.cases(churn), ]

cols_recode1 <- c(10:15)
for(i in 1:ncol(churn[,cols_recode1]))
  {
```

```
  churn[,cols_recode1][,i] <- as.factor(mapvalues
        (churn[,cols_recode1][,i], from =c("No internet service"),to=c("No")))
}

churn$MultipleLines <- as.factor(mapvalues(churn$MultipleLines,
                      from=c("No phone service"),to=c("No")))


min(churn$tenure); max(churn$tenure)
```

```
## [1] 1
```

```
## [1] 72
```

Data Cleaning

customerID has no significance so we are dropping this column

```
churn <- subset(churn, select = -customerID)
head(churn)
```

```
##   gender SeniorCitizen Partner Dependents tenure PhoneService MultipleLines
## 1 Female             0     Yes         No      1           No            No
## 2   Male             0      No         No     34          Yes            No
## 3   Male             0      No         No      2          Yes            No
## 4   Male             0      No         No     45           No            No
## 5 Female             0      No         No      2          Yes            No
## 6 Female             0      No         No      8          Yes           Yes
##   InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport
## 1             DSL             No          Yes               No          No
## 2             DSL            Yes           No              Yes          No
## 3             DSL            Yes          Yes               No          No
## 4             DSL            Yes           No              Yes         Yes
## 5     Fiber optic             No           No               No          No
## 6     Fiber optic             No           No              Yes          No
##   StreamingTV StreamingMovies       Contract PaperlessBilling
## 1          No              No Month-to-month              Yes
## 2          No              No       One year               No
## 3          No              No Month-to-month              Yes
## 4          No              No       One year               No
## 5          No              No Month-to-month              Yes
## 6         Yes             Yes Month-to-month              Yes
##               PaymentMethod MonthlyCharges TotalCharges Churn
## 1          Electronic check          29.85        29.85    No
## 2              Mailed check          56.95      1889.50    No
## 3              Mailed check          53.85       108.15   Yes
## 4 Bank transfer (automatic)          42.30      1840.75    No
## 5          Electronic check          70.70       151.65   Yes
## 6          Electronic check          99.65       820.50   Yes
```

```
# Handling Missing, Duplicate Data and null data
sum(duplicated(churn))
```

```
## [1] 22
```

```r
# There are 17 Duplicate rows & 11 missing values in TotalCharges Column.
# The number of Duplicate rows & missing values is relatively small compared
# to the size of the dataset. Therefore, I have decided to remove these missing
# values from our analysis.

datc <- churn %>%
        na.omit(churn) %>%
        distinct()
sum(duplicated(datc))
```

```
## [1] 0
```

Checking Data structure

```r
summary(churn)
```

```
##     gender          SeniorCitizen      Partner           Dependents
##  Length:7032        Min.   :0.0000   Length:7032        Length:7032
##  Class :character   1st Qu.:0.0000   Class :character   Class :character
##  Mode  :character   Median :0.0000   Mode  :character   Mode  :character
##                     Mean   :0.1624
##                     3rd Qu.:0.0000
##                     Max.   :1.0000
##      tenure      PhoneService       MultipleLines InternetService
##  Min.   : 1.00   Length:7032        No :4065      Length:7032
##  1st Qu.: 9.00   Class :character   Yes:2967      Class :character
##  Median :29.00   Mode  :character                 Mode  :character
##  Mean   :32.42
##  3rd Qu.:55.00
##  Max.   :72.00
##  OnlineSecurity OnlineBackup DeviceProtection TechSupport StreamingTV
##  No :5017       No :4607     No :4614         No :4992    No :4329
##  Yes:2015       Yes:2425     Yes:2418         Yes:2040    Yes:2703
##
##
##
##
##  StreamingMovies   Contract         PaperlessBilling   PaymentMethod
##  No :4301        Length:7032        Length:7032        Length:7032
##  Yes:2731        Class :character   Class :character   Class :character
##                  Mode  :character   Mode  :character   Mode  :character
##
##
##
##  MonthlyCharges    TotalCharges       Churn
##  Min.   : 18.25   Min.   :  18.8   Length:7032
##  1st Qu.: 35.59   1st Qu.: 401.4   Class :character
##  Median : 70.35   Median :1397.5   Mode  :character
##  Mean   : 64.80   Mean   :2283.3
##  3rd Qu.: 89.86   3rd Qu.:3794.7
##  Max.   :118.75   Max.   :8684.8
```

The “SeniorCitizen” variable in the dataset is currently coded as a binary. values of “0” or “1”.This coding scheme is not intuitive and can be difficult To make our analysis easier and more understandable,we can recode this intuitive labeling scheme, such as “yes” and “no”.

The “MultipleLines” variable is related to the “PhoneService” variable.

If “PhoneService” is set to “No”,then “MultipleLines” is also set to “No”.

To simplify our graphics and modeling,we can change the “No phone service”. “MultipleLines” variable to “No”.

Same goes with “No internet service”

```
datrecode <- churn %>%
mutate(SeniorCitizen = if_else(SeniorCitizen == 0,"No","Yes")) %>%
mutate(MultipleLines = if_else(MultipleLines == "No phone service","No",
                               MultipleLines)) %>%
mutate(across(starts_with("Online") | starts_with("Device") |
                starts_with("Tech") | starts_with("Streaming"),
              ~if_else(. == "No internet service", "No", .)))
head(datrecode)
```

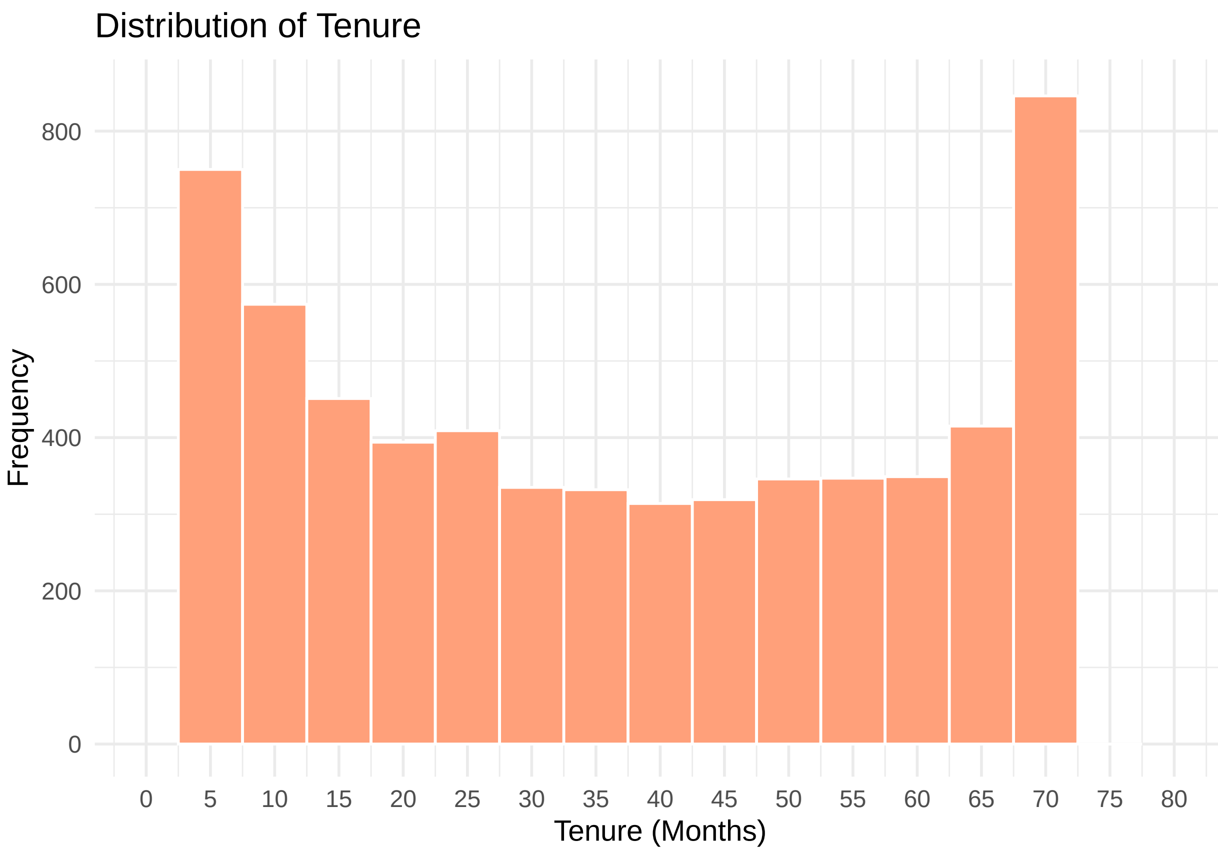
```
##   gender SeniorCitizen Partner Dependents tenure PhoneService MultipleLines
## 1 Female            No     Yes         No      1           No            No
## 2   Male            No      No         No     34          Yes            No
## 3   Male            No      No         No      2          Yes            No
## 4   Male            No      No         No     45           No            No
## 5 Female            No      No         No      2          Yes            No
## 6 Female            No      No         No      8          Yes           Yes
##   InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport
## 1             DSL             No          Yes               No          No
## 2             DSL            Yes           No              Yes          No
## 3             DSL            Yes          Yes               No          No
## 4             DSL            Yes           No              Yes         Yes
## 5     Fiber optic             No           No               No          No
## 6     Fiber optic             No           No              Yes          No
##   StreamingTV StreamingMovies       Contract PaperlessBilling
## 1          No              No Month-to-month              Yes
## 2          No              No       One year               No
## 3          No              No Month-to-month              Yes
## 4          No              No       One year               No
## 5          No              No Month-to-month              Yes
## 6         Yes             Yes Month-to-month              Yes
##               PaymentMethod MonthlyCharges TotalCharges Churn
## 1          Electronic check          29.85        29.85    No
## 2              Mailed check          56.95      1889.50    No
## 3              Mailed check          53.85       108.15   Yes
## 4 Bank transfer (automatic)          42.30      1840.75    No
## 5          Electronic check          70.70       151.65   Yes
## 6          Electronic check          99.65       820.50   Yes
```

Visualizations

Let's take a look at numerical Features

```r
# Histogram of Tenure:
p1 <- ggplot(datrecode, aes(x = tenure)) +
  geom_histogram(binwidth = 5, color = "white",fill="#FFA07A") +
  labs(title = "Distribution of Tenure",
       x = "Tenure (Months)",
       y = "Frequency") +
  scale_x_continuous(limits = c(0, 80), breaks = seq(0, 80, 5)) +
  theme_minimal()

p1
```

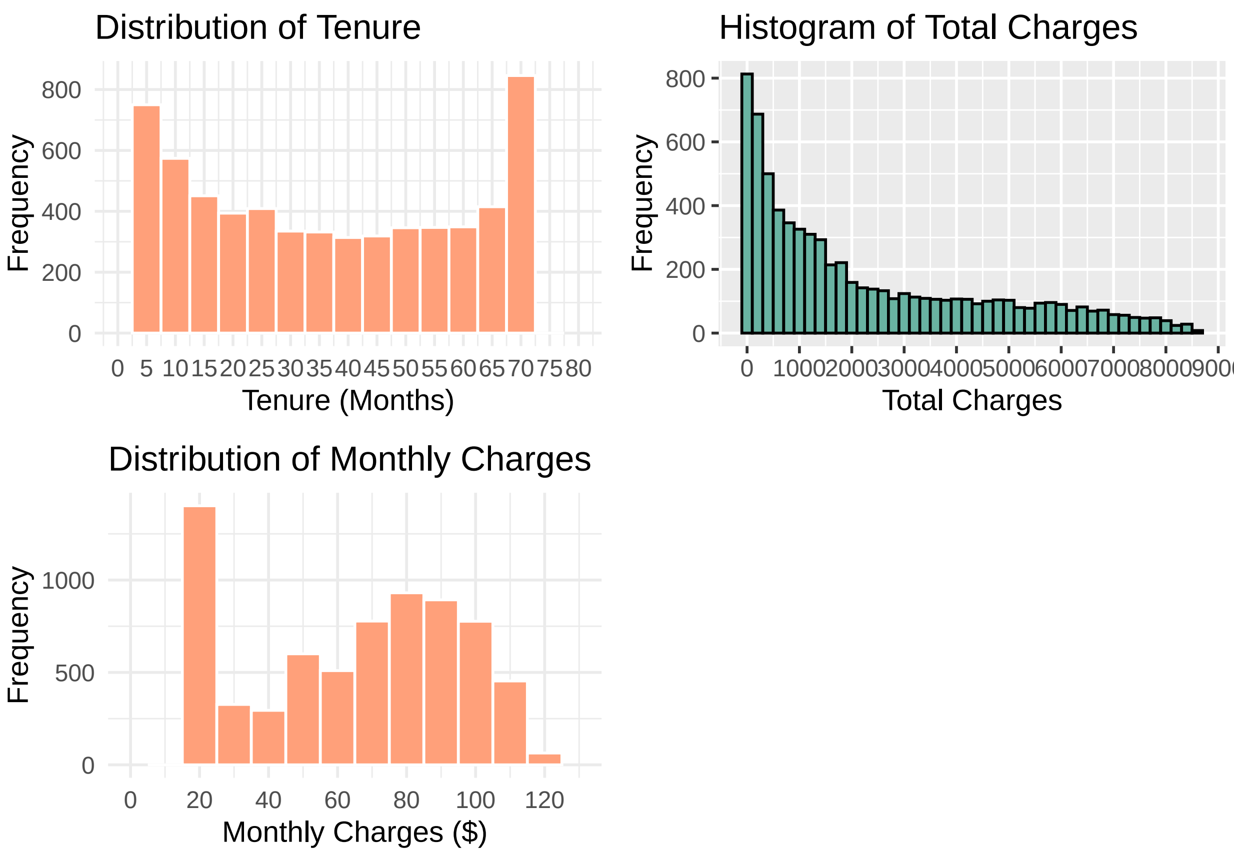
```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_bar()`).
```




```r
# Histogram of TotalCharges
p2 <- ggplot(datrecode, aes(x = TotalCharges)) +
  geom_histogram(binwidth = 200, fill = "#69b3a2", color = "black") +
  scale_x_continuous(breaks = seq(0, 9000, by = 1000)) +
  labs(x = "Total Charges", y = "Frequency",
       title = "Histogram of Total Charges")

# Histogram of MonthlyCharges
p3 <- ggplot(data = datrecode, aes(x = MonthlyCharges)) +
  geom_histogram(binwidth = 10, color = "white", fill = "#FFA07A") +
```

```
  labs(title = "Distribution of Monthly Charges",
       x = "Monthly Charges ($)",
       y = "Frequency") +
  scale_x_continuous(limits = c(0, 130), breaks = seq(0, 150, 20)) +
  theme_minimal()

grid.arrange(p1, p2, p3, ncol = 2)
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_bar()`).
## Removed 2 rows containing missing values or values outside the scale range
## (`geom_bar()`).
```



Let’s take a look at Categorical Features

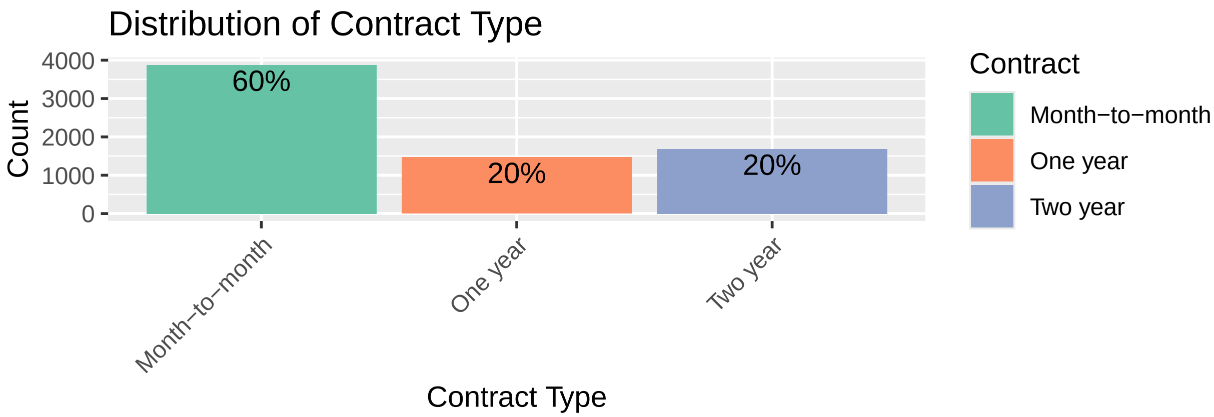
Bar graph of Contract

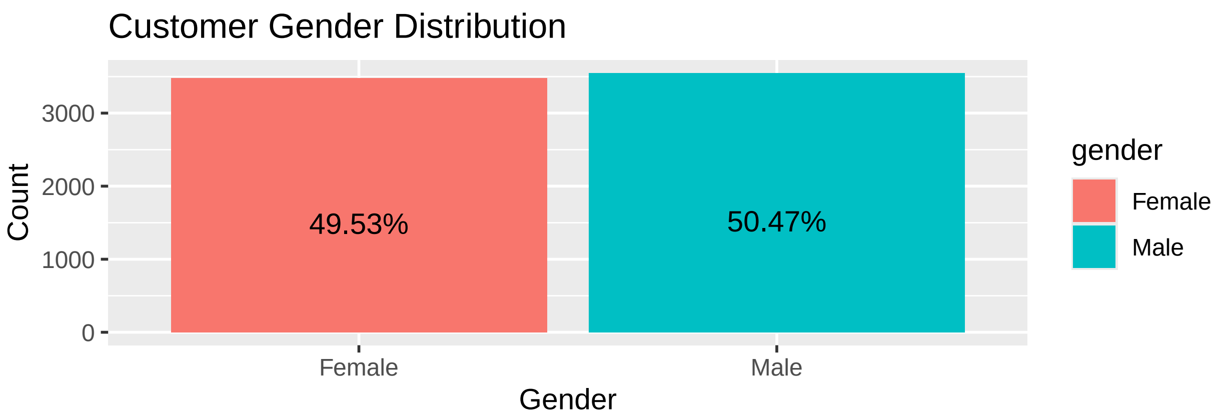
```
p9 <-  ggplot(datrecode, aes(x = Contract, fill = Contract)) +
geom_bar() +geom_text(aes(y = ..count.. -400,
                          label = paste0(round(prop.table(..count..),
    1)*100, "%")), stat = "count")+labs(title = "Distribution of Contract Type",
x = "Contract Type",
y = "Count") +
scale_fill_brewer(palette = "Set2") +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Bar graph of gender

```
colors <- c("#0072B2", "#E69F00")

p10 <- ggplot(data = datrecode, aes(x = gender))+
  geom_bar(aes(fill = gender))+
geom_text(aes(y = ..count.. -500,
    label = paste0(round(prop.table(..count..),4)* 100, '%')),
  stat = "count",
position =
  position_stack(vjust = 0.5))+
  xlab("Gender")+
  ylab("Count")+
  ggtitle("Customer Gender Distribution")+
  theme()

# Arrange plots in a grid
grid.arrange(p9, p10)
```

```
## Warning: The dot-dot notation (`..count..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(count)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

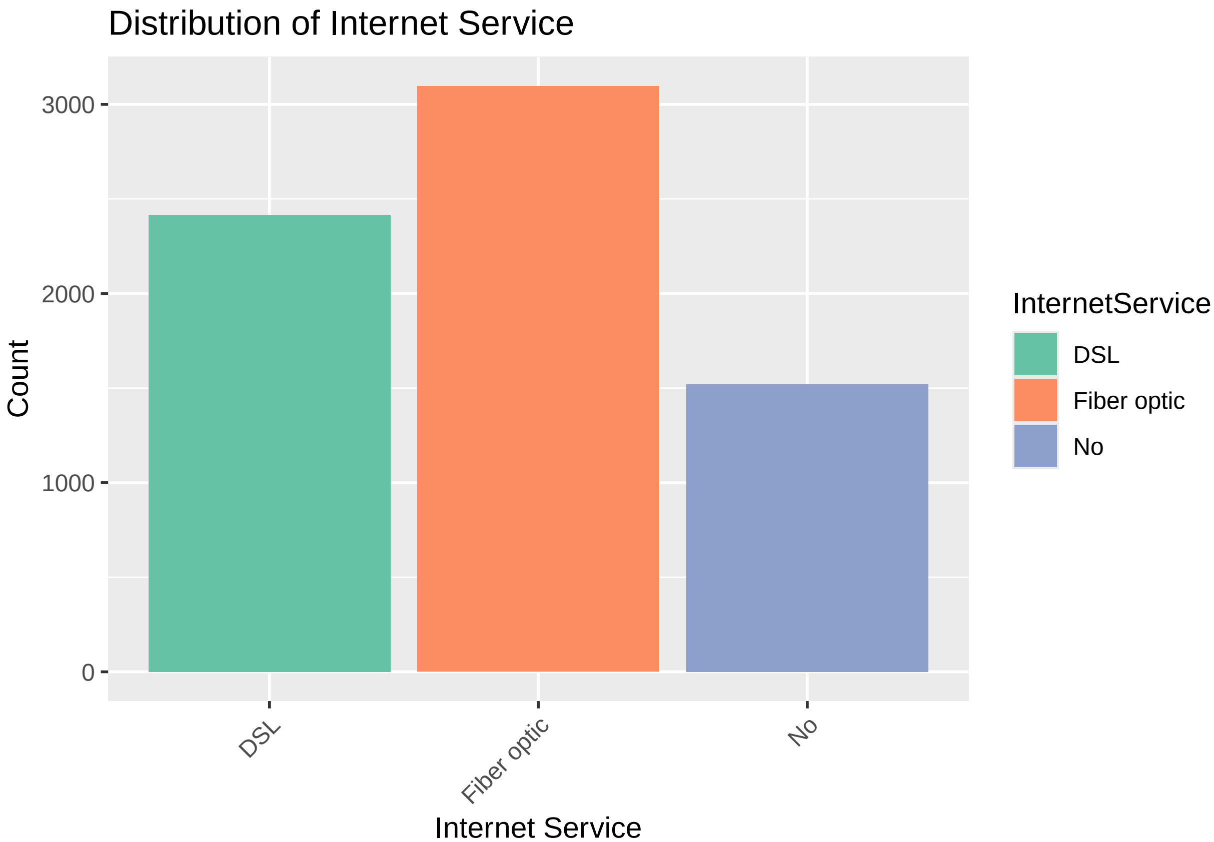




#

Bar graph of InternetService

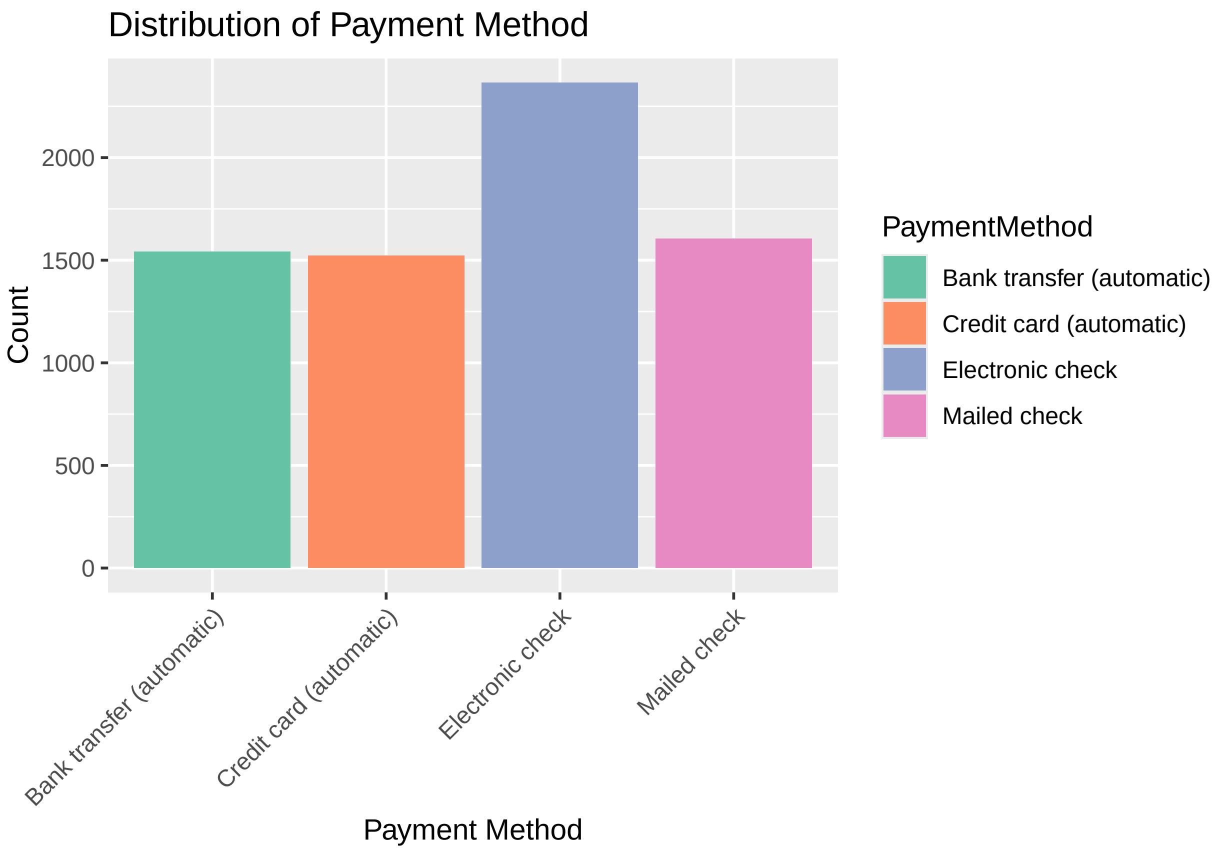
```
p11 <- ggplot(datrecode, aes(x = InternetService, fill = InternetService)) +
  geom_bar() +
  labs(title = "Distribution of Internet Service",
       x = "Internet Service",
       y = "Count") +
  scale_fill_brewer(palette = "Set2") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

p11
```



#

Bar graph of PaymentMethod

```
p12 <- ggplot(datrecode, aes(x = PaymentMethod, fill = PaymentMethod)) +
  geom_bar() +
  labs(title = "Distribution of Payment Method",
       x = "Payment Method",
       y = "Count") +
  scale_fill_brewer(palette = "Set2") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

p12
```

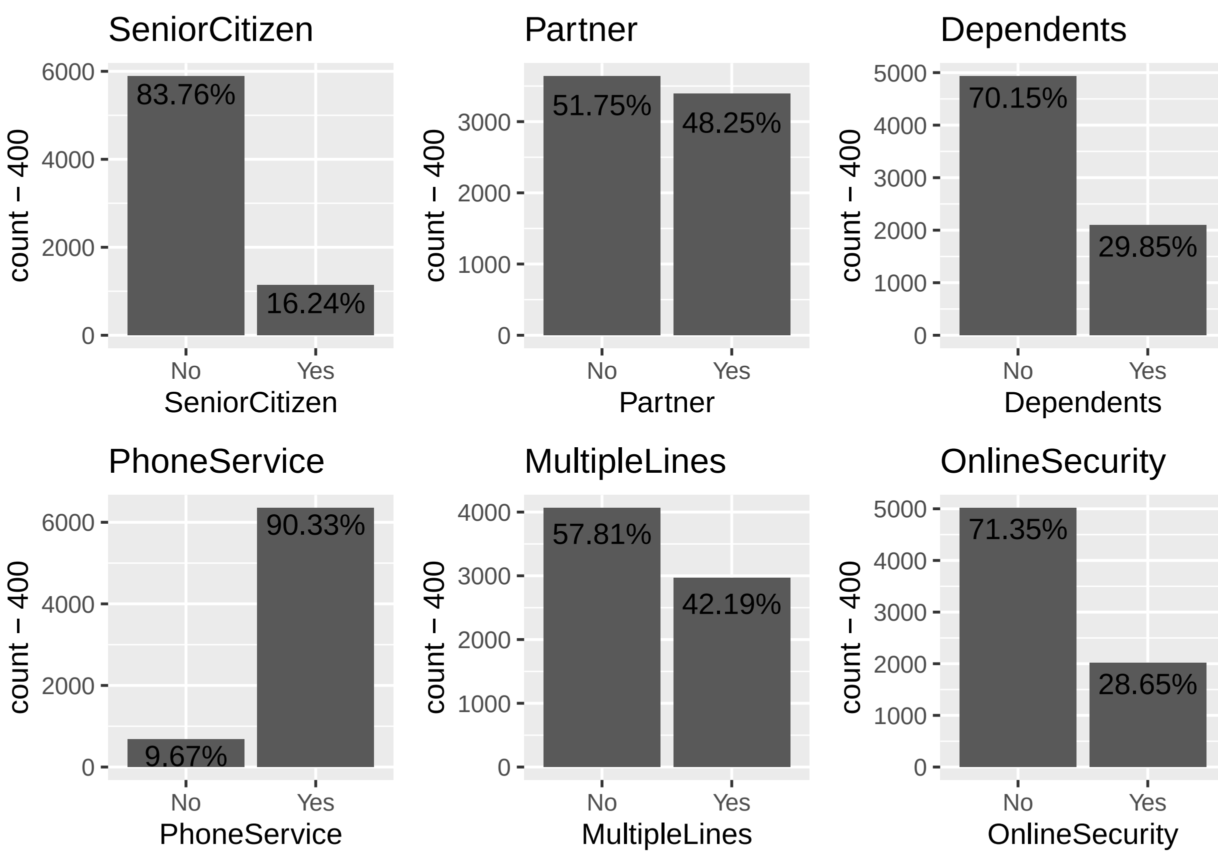


Plotting Bar graphs of SeniorCitizen, Partner, Dependents, PhoneService,

MultipleLines and OnlineSecurity.

```
p14 <- ggplot(datrecode, aes(x = SeniorCitizen)) + geom_bar() +
  geom_text(aes(y = ..count.. -400,
                    label = paste0(round(prop.table(..count..),4)*100, '%')),
                      stat = 'count')+ ggtitle("SeniorCitizen")

p15 <- ggplot(datrecode, aes(x = Partner)) + geom_bar() +
  geom_text(aes(y = ..count.. -400,
                    label = paste0(round(prop.table(..count..),4)*100, '%')),
                      stat = 'count')+ ggtitle("Partner")

p16 <- ggplot(datrecode, aes(x = Dependents)) + geom_bar() +
  geom_text(aes(y = ..count.. -400,
                label = paste0(round(prop.table(..count..),4)*100, '%')),
                      stat = 'count')+ ggtitle("Dependents")

p17 <- ggplot(datrecode, aes(x = PhoneService)) + geom_bar() +
  geom_text(aes(y = ..count.. -400,
                    label = paste0(round(prop.table(..count..),4)*100, '%')),
                      stat = 'count')+ ggtitle("PhoneService")
```

```r
p18 <- ggplot(datrecode, aes(x = MultipleLines)) + geom_bar() +
  geom_text(aes(y = ..count.. -400,
                label = paste0(round(prop.table(..count..),4)*100, '%')),
                   stat = 'count')+ ggtitle("MultipleLines")

p19 <- ggplot(datrecode, aes(x = OnlineSecurity)) + geom_bar() +
  geom_text(aes(y = ..count.. -400,
                  label = paste0(round(prop.table(..count..),4)*100, '%')),
                   stat = 'count')+ ggtitle("OnlineSecurity")
# Arrange plots in a grid
grid.arrange(p14,p15,p16,p17,p18,p19, nrow = 2, ncol = 3)
```

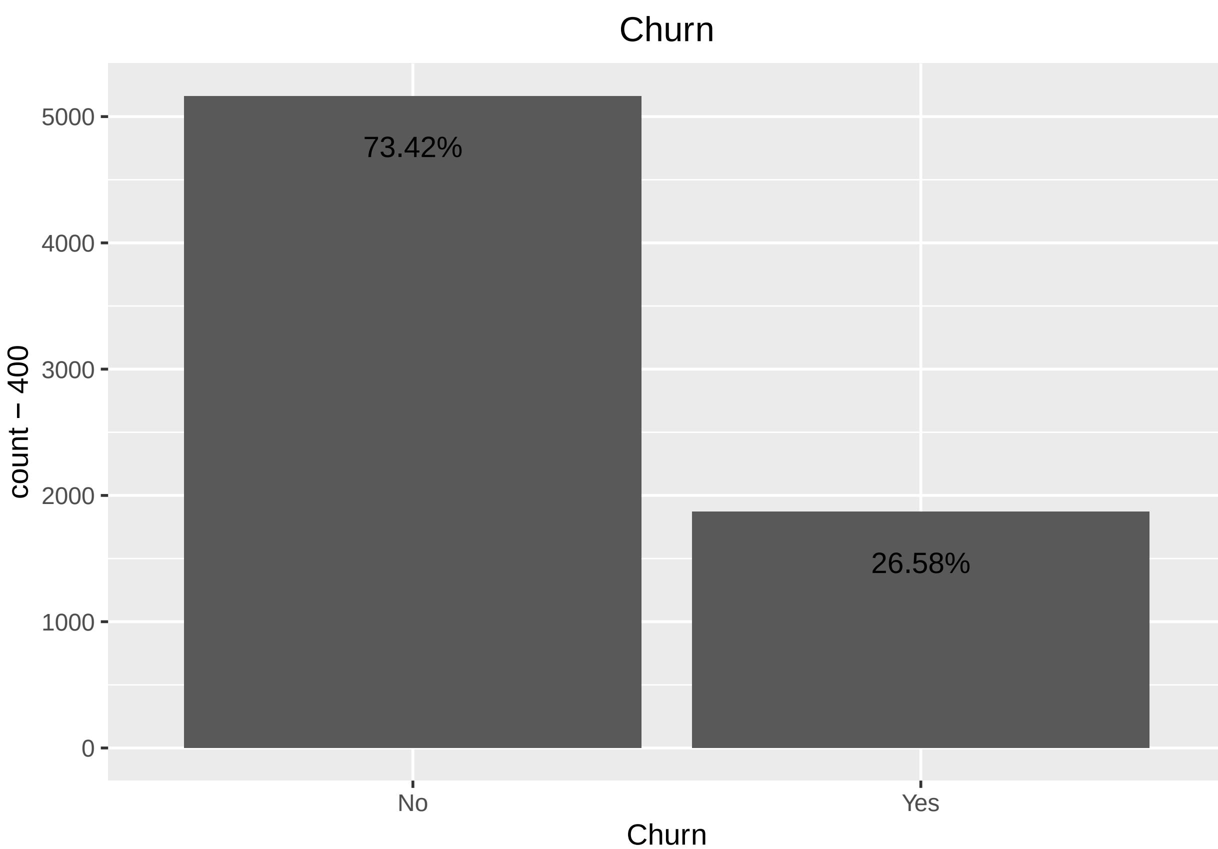


Bar graphs of OnlinBackup,DeviceProtection,TechSupport,StreamingTV, StreamingMovies and paperLessbill.

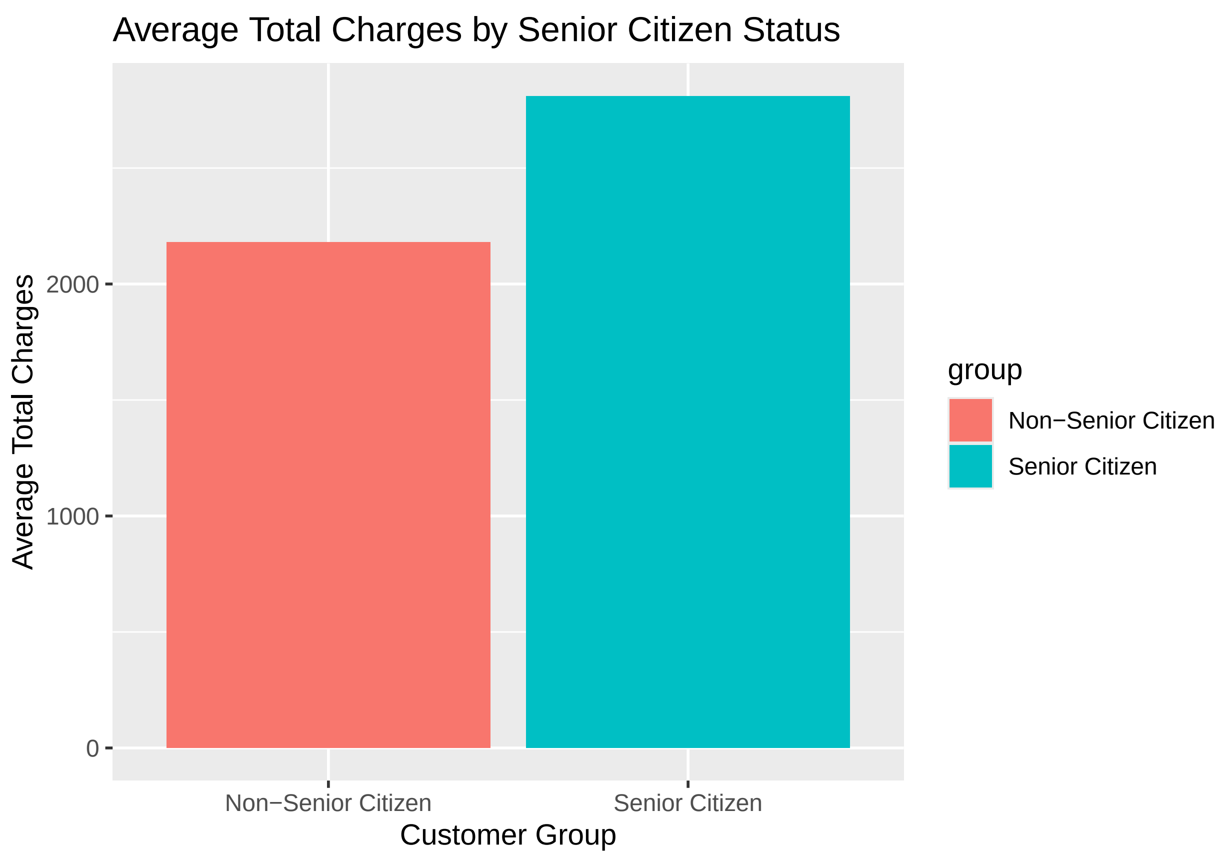
```r
p20 <- ggplot(datrecode, aes(x = OnlineBackup)) + geom_bar() +
  geom_text(aes(y = ..count.. -1000,
                label = paste0(round(prop.table(..count..),4)*100, '%')),
                stat = 'count')+ ggtitle("OnlineBackup")

p21 <- ggplot(datrecode, aes(x = DeviceProtection)) + geom_bar()+
  geom_text(aes(y= ..count.. -400,
                label = paste0(round(prop.table(..count..),4)*100, "%")),
                stat = "count") +ggtitle("DeviceProtection")
```

```r
p22 <- ggplot(datrecode, aes(x = TechSupport)) + geom_bar()+
   geom_text(aes(y= ..count.. -400,
                       label = paste0(round(prop.table(..count..),4)*100, "%")),
                       stat = "count")  +ggtitle("TechSupport")

p23 <- ggplot(datrecode, aes(x = StreamingTV)) + geom_bar() +
  geom_text(aes(y= ..count.. -400,
            label = paste0(round(prop.table(..count..),4)*100, "%")),
            stat = "count") + ggtitle("StreamingTV")

p24 <- ggplot(datrecode, aes(x = StreamingMovies)) +
  geom_bar() +geom_text(aes(y= ..count.. -400,
            label = paste0(round(prop.table(..count..),4)*100, "%")),
            stat = "count") + ggtitle("StreamingMovies")

p25 <- ggplot(datrecode, aes(x = PaperlessBilling)) +
  geom_bar() +geom_text(aes(y= ..count.. -400,
                       label = paste0(round(prop.table(..count..),4)*100, "%")),
                         stat = "count") + ggtitle("PaperlessBilling")
```

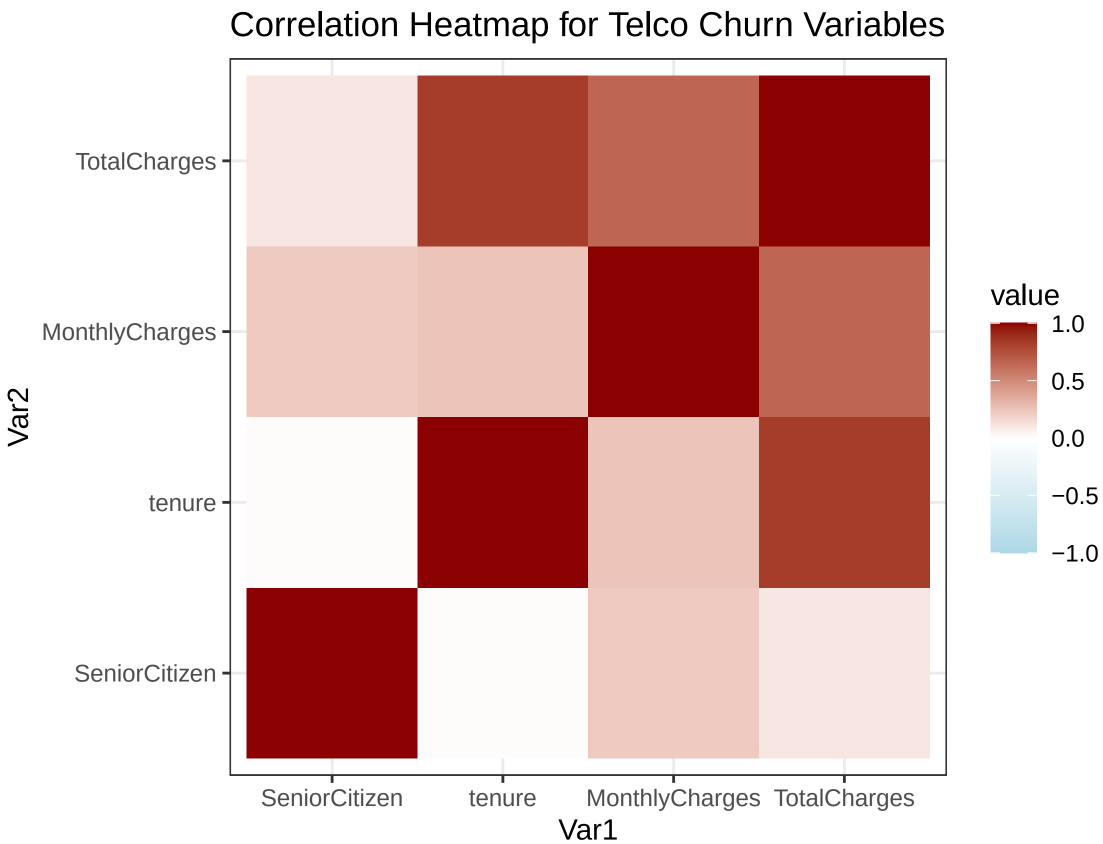
```r
# How many people has chuned from our data
p13 <- ggplot(datrecode, aes(x = Churn)) +
  geom_bar()+ geom_text(aes(y = ..count.. -400,
                         label = paste0(round(prop.table(..count..),4)*100,'%')),
                         stat = 'count') + ggtitle("Churn")+
  theme(plot.title = element_text(hjust = .5))

p13
```




```r
# Calculate average total charges for senior citizens

avg_total_charges_senior <- mean(churn[churn$SeniorCitizen == 1, ]$TotalCharges)

# Calculate average total charges for non-senior citizens

avg_total_charges_non_senior<-mean(churn[churn$SeniorCitizen==0, ]$TotalCharges)

# Create a bar chart for visualization

ggplot(data.frame(group = c("Senior Citizen", "Non-Senior Citizen"),
      avg_charges = c(avg_total_charges_senior, avg_total_charges_non_senior)),
       aes(x = group, y = avg_charges, fill = group)) +
  geom_bar(stat = "identity") +

  labs(title = "Average Total Charges by Senior Citizen Status",
       x = "Customer Group",
       y = "Average Total Charges")
```

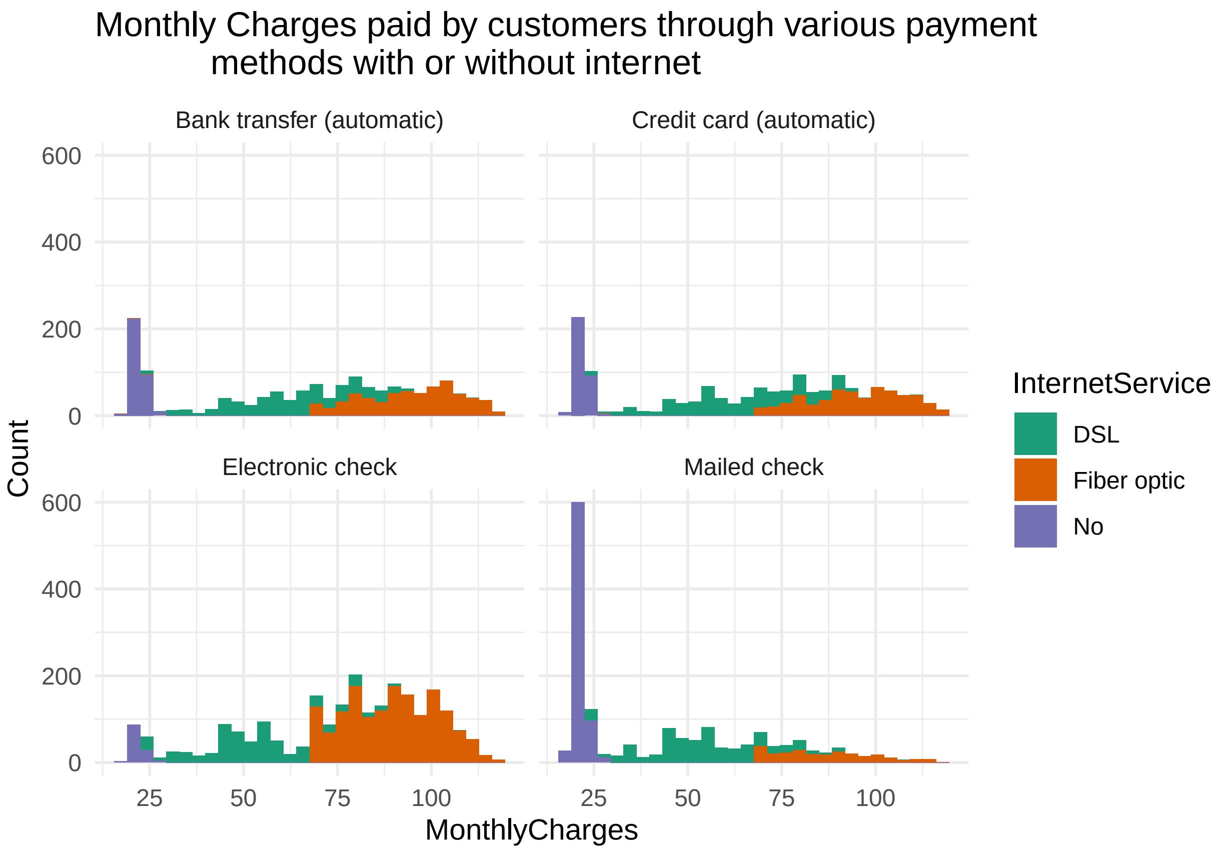



```
correlation_matrix <- cor(churn[, sapply(churn, is.numeric)])

ggplot(melt(correlation_matrix), aes(x = Var1, y = Var2, fill = value)) +
geom_tile() +
scale_fill_gradient2(low="lightblue",high="darkred",midpoint=0,limits=c(-1,1)) +
  theme_bw() +
  labs(title = "Correlation Heatmap for Telco Churn Variables") +
  coord_fixed()
```

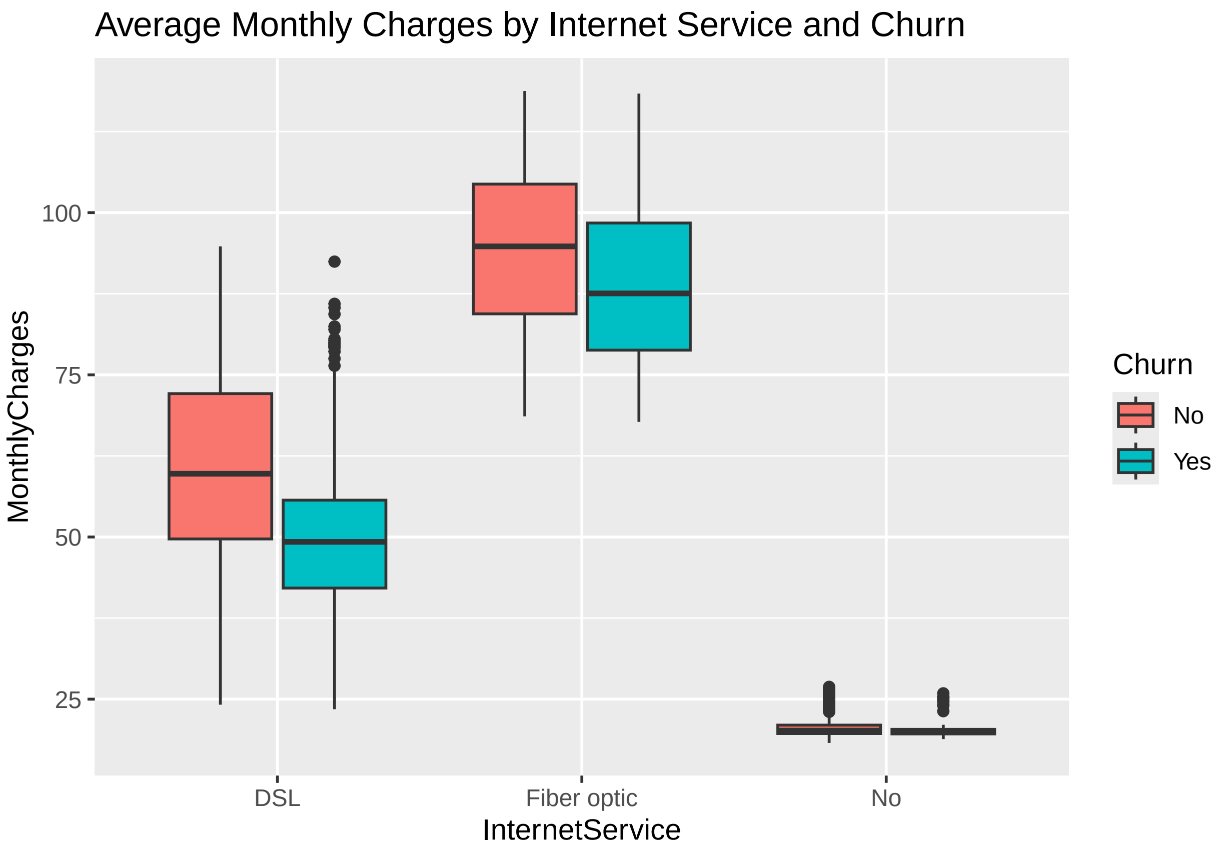



```
churn %>% ggplot(aes(x = MonthlyCharges, fill = InternetService)) +
    facet_wrap(~PaymentMethod) + theme_minimal() +
    scale_fill_brewer(palette="Dark2") +
    ggtitle(label = "Monthly Charges paid by customers through various payment
            methods with or without internet") +
    ylab(label = "Count") + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```r
# Calculate average monthly charges per internet service type
avg_charges_by_service <- datrecode %>%
  group_by(InternetService) %>%
  summarize(avg_monthly_charge = mean(MonthlyCharges))

# Visualize distribution of average monthly charges by churn status
ggplot(datrecode, aes(x = InternetService, y = MonthlyCharges, fill = Churn)) +
  geom_boxplot() +
  labs(title = "Average Monthly Charges by Internet Service and Churn")
```



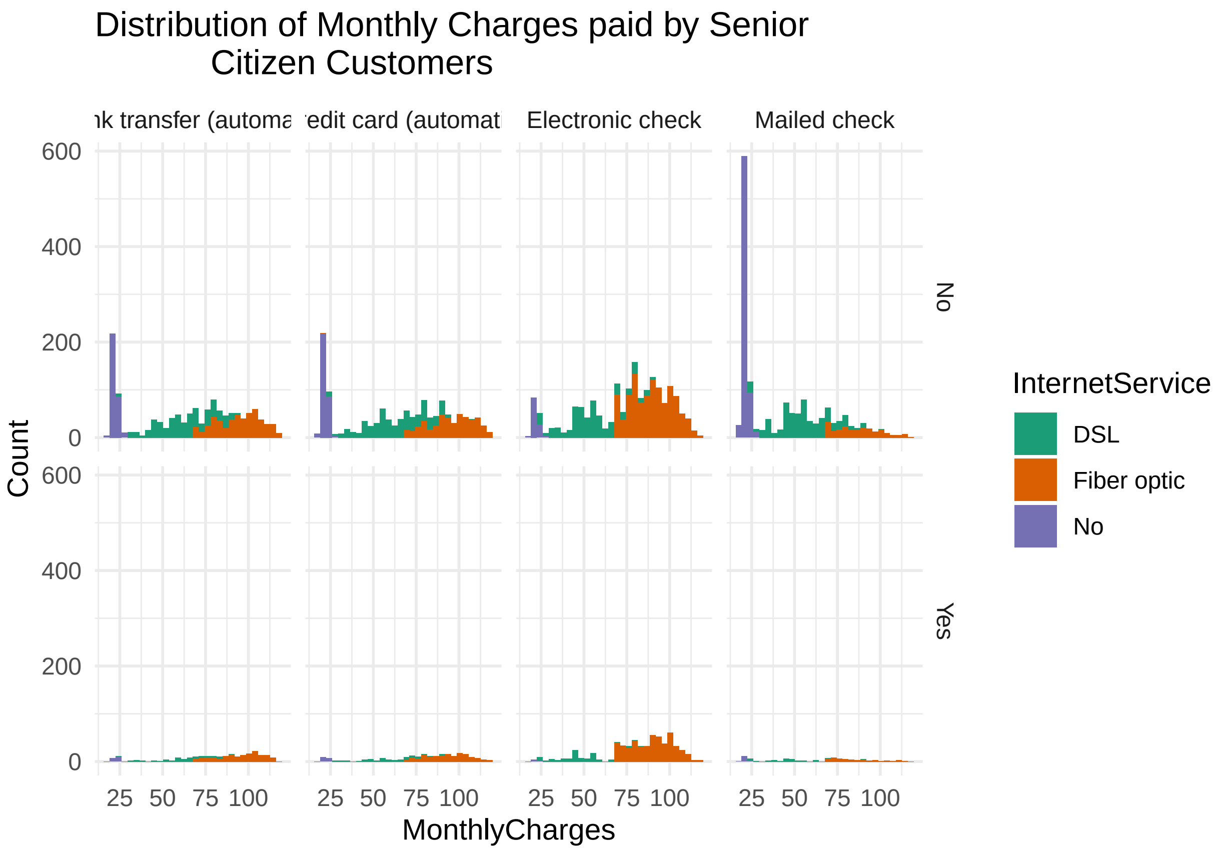

```
# Correlation analysis between average monthly charges and churn
cor(datrecode$MonthlyCharges, as.numeric(datrecode$Churn))
```

```
## Warning in is.data.frame(y): NAs introduced by coercion
```

```
## [1] NA
```

```
# Calculate Distribution of Monthly Charges paid by Senior Citizen Customers
ggplot(datrecode,aes(x = MonthlyCharges, fill = InternetService)) +
    geom_histogram() +
    facet_grid(SeniorCitizen~PaymentMethod) +
    theme_minimal() +
    scale_fill_brewer(palette="Dark2") +
    ggtitle(label = "Distribution of Monthly Charges paid by Senior
            Citizen Customers") +
    ylab(label = "Count")
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```


Distribution of Monthly Charges paid by Senior Citizen Customers
nk transfer (automa
redit card (automat
Electronic check
Mailed check
600
400
200
0
600
400
200
0
Count
No
Yes
25 50 75 100
25 50 75 100
25 50 75 100
25 50 75 100
MonthlyCharges
InternetService
DSL
Fiber optic
No